

```

/*-----*
|  CONFUNDING DEMO (covariance pattern / marginal model)
|  No random effects. Uses PROC MIXED with REPEATED.
|
|  Variables:
|  id      = subject ID
|  time    = visit time (0,1,2,3,4)  (treated as CLASS)
|  y       = outcome
|  A       = exposure (0/1), baseline
|  age     = baseline confounder (associated with A and y)
|  biom    = time-varying covariate affected by prior A (mediator-ish)
|  z       = precision covariate (baseline-only here)
/*-----*/

```

```

proc print data = demo_long (obs=20);
run;

```

Obs	id	age	A	z	time	biom	y
1	1	51.1130	1	-1.77231	0	-0.53577	24.3368
2	1	51.1130	1	-1.77231	1	-1.23809	25.7489
3	1	51.1130	1	-1.77231	2	0.11968	24.8903
4	1	51.1130	1	-1.77231	3	0.47853	32.5064
5	1	51.1130	1	-1.77231	4	1.71218	30.9145
6	2	38.0998	1	0.61437	0	0.73368	25.7812
7	2	38.0998	1	0.61437	1	1.35669	30.8212
8	2	38.0998	1	0.61437	2	2.47425	33.4281
9	2	38.0998	1	0.61437	3	0.18479	31.7852
10	2	38.0998	1	0.61437	4	0.08670	30.5793
11	3	57.4642	1	-0.35376	0	0.40352	24.5160
12	3	57.4642	1	-0.35376	1	2.62107	28.4818
13	3	57.4642	1	-0.35376	2	3.25722	32.2862
14	3	57.4642	1	-0.35376	3	3.61449	35.8133

Obs	id	age	A	z	time	biom	y
15	3	57.4642	1	-0.35376	4	4.02203	34.3625
16	4	55.1957	0	-2.13380	0	0.07050	22.9312
17	4	55.1957	0	-2.13380	1	-1.14543	20.0592
18	4	55.1957	0	-2.13380	2	-0.21004	22.3938
19	4	55.1957	0	-2.13380	3	0.58952	26.3002
20	4	55.1957	0	-2.13380	4	-0.42391	23.2728

title "Spaghetti plots by exposure group with mean trajectories";

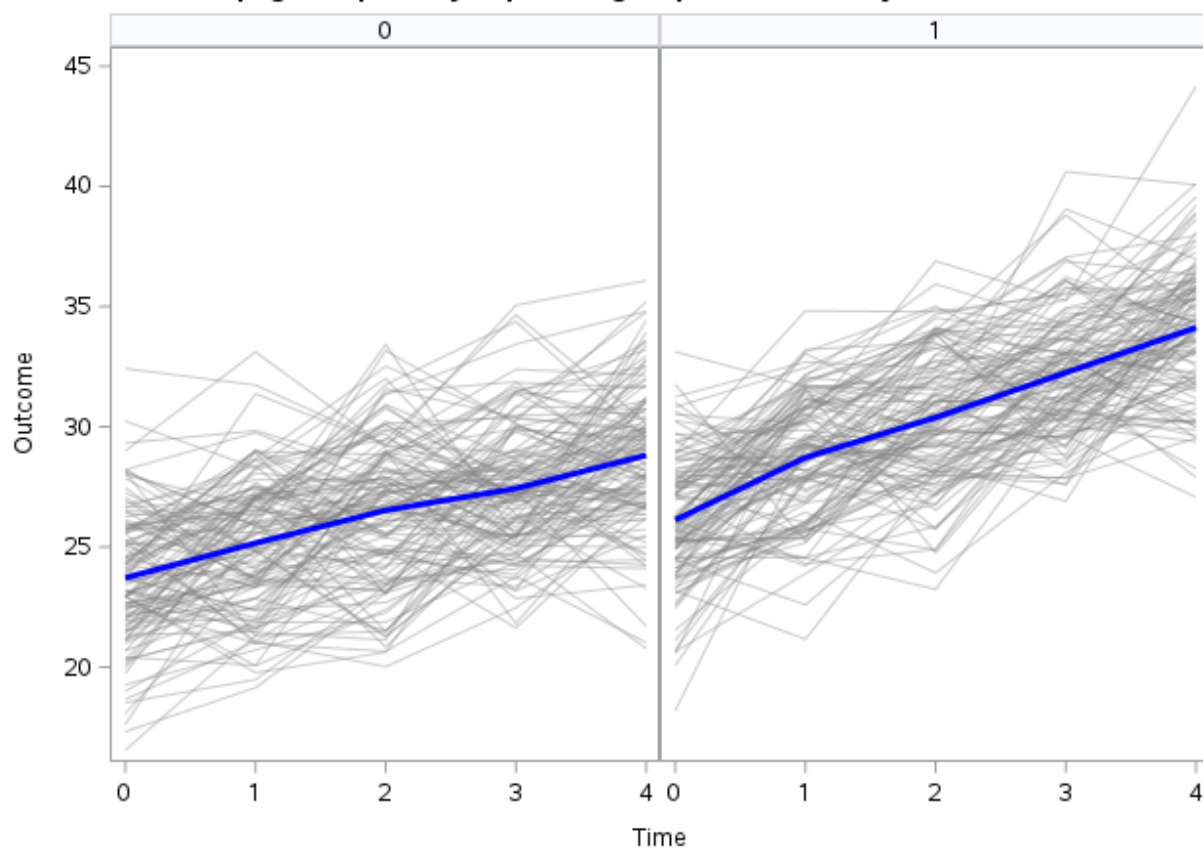
```
/* Step 1: Compute group-specific means at each time */
proc sort data=demo_long;
  by A time;
run;
```

```
proc means data=demo_long noprint;
  by A time;
  var y;
  output out=demo_means mean=mean_y;
run;
```

```
/* Step 2: Stack individual data and means */
data demo_plot;
  set demo_long demo_means;
run;
```

```
/* Step 3: Panel plot with mean overlay */
proc sgpanel data=demo_plot;
  panelby A / columns=2 novarname;
  series x=time y=y / group=id
    lineattrs=(color=gray pattern=1)
    transparency=0.6;
  series x=time y=mean_y /
    lineattrs=(color=blue thickness=3);
  colaxis values=(0 1 2 3 4) label="Time";
  rowaxis label="Outcome";
run;
```

Spaghetti plots by exposure group with mean trajectories



```

/*=====
|  A) COVARIANCE SELECTION USING REML (fixed mean model)
/*=====*/

title "A) Covariance selection using REML (mean fixed: A + time + A*time)";

/* Fit the same mean model under several covariance structures */
ods exclude all;

/* CS */
ods output FitStatistics=Fit_REML_CS;
proc mixed data=demo_long method=reml;
  class id time A;
  model y = A time A*time / solution ddfm=kr;
  repeated time / subject=id type=cs;
run;

/* AR(1) */
ods output FitStatistics=Fit_REML_AR1;
proc mixed data=demo_long method=reml;
  class id time A;
  model y = A time A*time / solution ddfm=kr;
  repeated time / subject=id type=ar(1);
run;

/* UN */
ods output FitStatistics=Fit_REML_UN;
proc mixed data=demo_long method=reml;
  class id time A;
  model y = A time A*time / solution ddfm=kr;
  repeated time / subject=id type=un;
run;

ods exclude none;

/* Combine fit stats for display */
data Fit_REML_All;
  length cov $12;
  set Fit_REML_CS(in=a) Fit_REML_AR1(in=b) Fit_REML_UN(in=c);
  if a then cov="CS";
  if b then cov="AR(1)";
  if c then cov="UN";
run;

data Fit_REML_All2;
  set Fit_REML_All;
  length stat $20;
  if index(upcase(descr), "-2 RES LOG LIKELIHOOD")>0 then stat="-2ResLL";
  else if index(upcase(descr), "AIC")>0 and index(upcase(descr), "AICC")=0 then
stat="AIC";
  else if index(upcase(descr), "BIC")>0 then stat="BIC";
  else delete;
run;

proc sort data=Fit_REML_All2;
  by cov stat;
run;

```

```

proc transpose data=Fit_REML_All2 out=Fit_REML_Wide(drop=_name_);
  by cov;
  id stat;
  var value;
run;

proc print data=Fit_REML_Wide noobs;
  title "REML fit statistics (wide format)";
run;

```

REML fit statistics (wide format)

cov	-2ResLL	AIC	BIC
AR(1)	5888.7	5892.7	5899.8
CS	5811.7	5815.7	5822.8
UN	5798.3	5828.3	5881.2

```

/*=====
|  B) CONFOUNDER SELECTION USING ML (cov fixed to CS)
|    - Use ML because fixed effects differ across models.
|    - Demonstrate change-in-estimate on one A*time term.
|    - Also show LRT/AIC/BIC if you want.
/*=====*/

title "Baseline covariate comparisons by exposure group (time = 0)";

/* Age */
proc ttest data=demo_long(where=(time=0));
  class A;
  var age  z biom;
run;

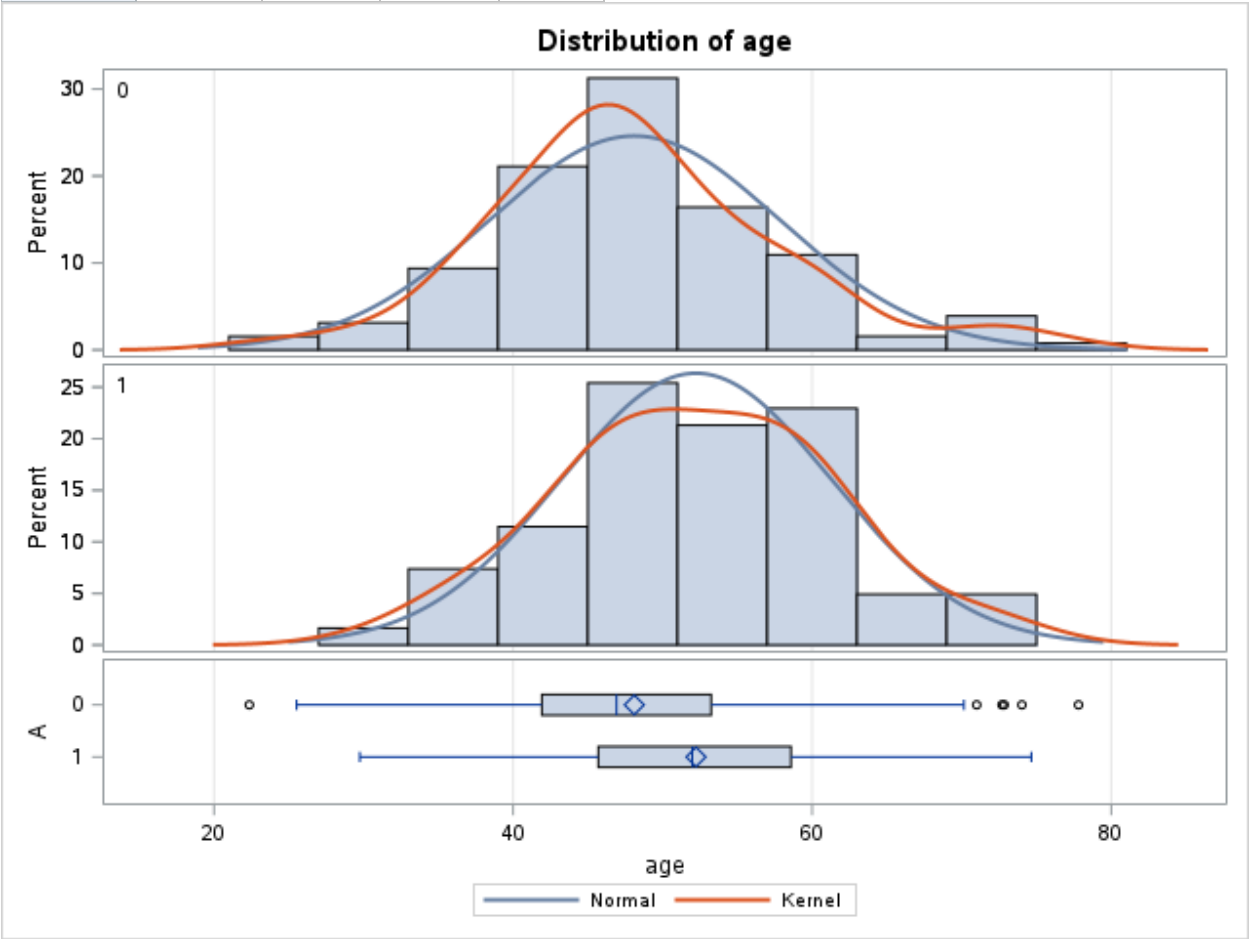
title;

```

Baseline covariate comparisons by exposure group (time = 0) The TTEST Procedure

Variable: age							
A	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		128	48.1074	9.7293	0.8600	22.3624	77.8129
1		122	52.2326	9.0836	0.8224	29.7504	74.6588
Diff (1-2)	Pooled		-4.1252	9.4198	1.1919		
Diff (1-2)	Satterthwaite		-4.1252		1.1899		
A	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		48.1074	46.4057	49.8091	9.7293	8.6658	11.0928
1		52.2326	50.6044	53.8607	9.0836	8.0691	10.3922
Diff (1-2)	Pooled	-4.1252	-6.4727	-1.7777	9.4198	8.6587	10.3287
Diff (1-2)	Satterthwaite	-4.1252	-6.4688	-1.7816			
Method		Variances	DF	t Value	Pr > t		
Pooled		Equal	248	-3.46	0.0006		
Satterthwaite		Unequal	247.9	-3.47	0.0006		

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	127	121	1.15	0.4468

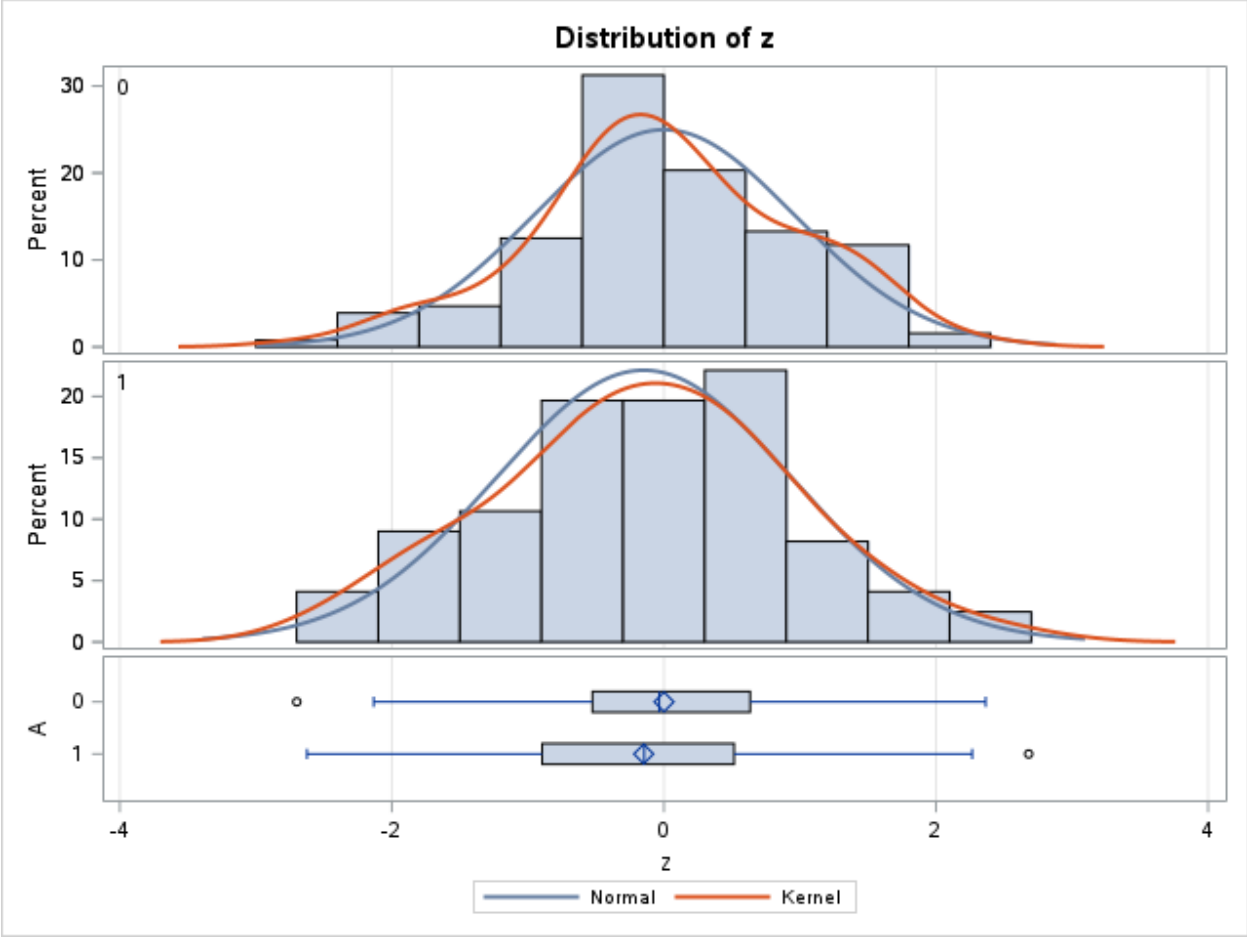


Variable: z							
A	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		128	0.000713	0.9584	0.0847	-2.6989	2.3653
1		122	-0.1488	1.0812	0.0979	-2.6238	2.6814
Diff (1-2)	Pooled		0.1495	1.0202	0.1291		
Diff (1-2)	Satterthwaite		0.1495		0.1295		

A	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		0.000713	-0.1669	0.1683	0.9584	0.8537	1.0928
1		-0.1488	-0.3426	0.0450	1.0812	0.9605	1.2370
Diff (1-2)	Pooled	0.1495	-0.1048	0.4037	1.0202	0.9378	1.1186
Diff (1-2)	Satterthwaite	0.1495	-0.1055	0.4045			

Method		Variances	DF	t Value	Pr > t
Pooled		Equal	248	1.16	0.2480
Satterthwaite		Unequal	241.21	1.15	0.2494

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	121	127	1.27	0.1802



Variable: biom

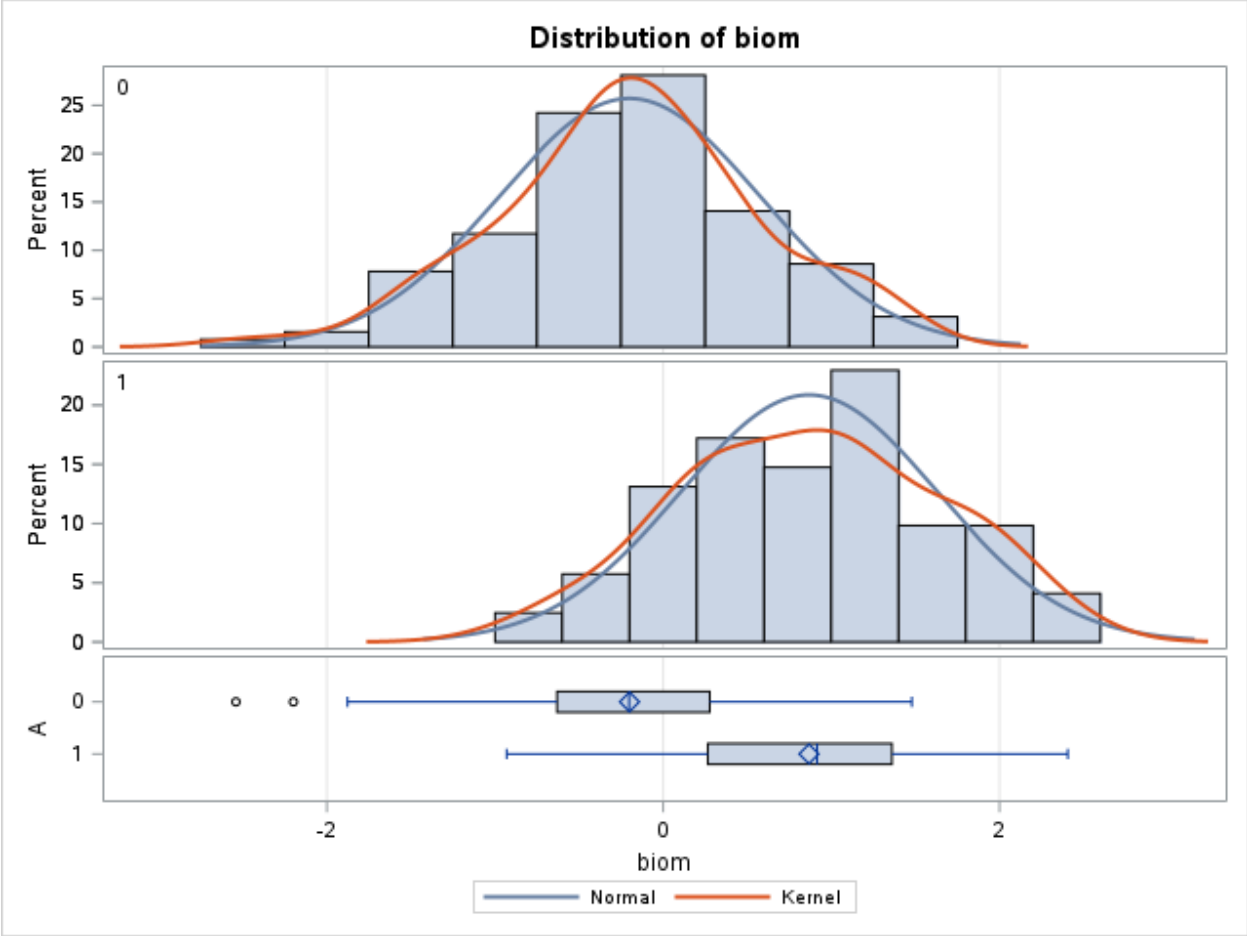
A	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		128	-0.2001	0.7758	0.0686	-2.5405	1.4786
1		122	0.8669	0.7648	0.0692	-0.9305	2.4061
Diff (1-2)	Pooled		-1.0670	0.7704	0.0975		
Diff (1-2)	Satterthwaite		-1.0670		0.0974		

A	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		-0.2001	-0.3358	-0.0644	0.7758	0.6910	0.8845
1		0.8669	0.7298	1.0039	0.7648	0.6793	0.8749
Diff (1-2)	Pooled	-1.0670	-1.2590	-0.8750	0.7704	0.7082	0.8448
Diff (1-2)	Satterthwaite	-1.0670	-1.2589	-0.8751			

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	248	-10.95	<.0001
Satterthwaite	Unequal	247.72	-10.95	<.0001

Equality of Variances

Method	Num DF	Den DF	F Value	Pr > F
Folded F	127	121	1.03	0.8745



```

/*-----*
|  Model 0: Base model (no covariates)
|  Mean: y = A + time + A*time
|  Covariance: CS
|  Method: ML (since we'll compare fixed effects across models)
|-----*/
title2 "Model 0: Base (A + time + A*time)";
proc mixed data=demo_long method=ml;
  class id;
  model y = A time A*time / solution;
  repeated / subject=id type=cs;

  /* Exposure effect at time=0: beta_A */
  estimate "Effect of A at time=0" A 1;

  /* Exposure effect at time=4: beta_A + 4*beta_{A*time} */
  estimate "Effect of A at time=4" A 1 A*time 4;
run;

```

Model 0: Base (A + time + A*time)

The Mixed Procedure

Model Information	
Data Set	WORK.DEMO_LONG
Dependent Variable	y
Covariance Structure	Compound Symmetry
Subject Effect	id
Estimation Method	ML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information

Class	Levels	Values
id	250	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165

Class Level Information																	
Class	Levels	Values															
		166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250															
Dimensions																	
Covariance Parameters		2															
Columns in X		4															
Columns in Z		0															
Subjects		250															
Max Obs per Subject		5															
Number of Observations																	
Number of Observations Read		1250															
Number of Observations Used		1250															
Number of Observations Not Used		0															
Iteration History																	
Iteration	Evaluations	-2 Log Like	Criterion														
0	1	6059.28259309															
1	1	5808.09719779	0.00000000														
Convergence criteria met.																	
Covariance Parameter Estimates																	
Cov Parm	Subject	Estimate															
CS	id	2.8605															

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
Residual		4.5994

Fit Statistics	
-2 Log Likelihood	5808.1
AIC (Smaller is Better)	5820.1
AICC (Smaller is Better)	5820.2
BIC (Smaller is Better)	5841.2

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
1	251.19	<.0001

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	23.8329	0.2095	248	113.74	<.0001
A	2.5803	0.3000	248	8.60	<.0001
time	1.2475	0.05994	998	20.81	<.0001
A*time	0.7057	0.08581	998	8.22	<.0001

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
A	1	248	74.00	<.0001
time	1	998	433.11	<.0001
A*time	1	998	67.63	<.0001

Estimates					
Label	Estimate	Standard Error	DF	t Value	Pr > t
Effect of A at time=0	2.5803	0.3000	248	8.60	<.0001
Effect of A at time=4	5.4030	0.3000	998	18.01	<.0001

```

/*-----*
|  Model 1: Add baseline confounder AGE (linear)
*-----*/
title2 "Model 1: + age (baseline confounder)";
proc mixed data=demo_long method=ml;
  class id;
  model y = A time A*time age / solution;
  repeated / subject=id type=cs;

  estimate "Effect of A at time=0" A 1;
  estimate "Effect of A at time=4" A 1 A*time 4;
run;

```

Model 1: + age (baseline confounder)

The Mixed Procedure

Model Information	
Data Set	WORK.DEMO_LONG
Dependent Variable	y
Covariance Structure	Compound Symmetry
Subject Effect	id
Estimation Method	ML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information

Class	Levels	Values
id	250	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250

Dimensions	
Covariance Parameters	2
Columns in X	5
Columns in Z	0
Subjects	250
Max Obs per Subject	5

Number of Observations	
Number of Observations Read	1250
Number of Observations Used	1250
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Log Like	Criterion
0	1	5937.77775554	
1	1	5757.63337580	0.00000000

Convergence criteria met.

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
CS	id	2.1695
Residual		4.5994

Fit Statistics	
-2 Log Likelihood	5757.6
AIC (Smaller is Better)	5771.6

Fit Statistics		
AICC (Smaller is Better)		5771.7
BIC (Smaller is Better)		5796.3
Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
1	180.14	<.0001

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	19.5705	0.6028	247	32.46	<.0001
A	2.2148	0.2851	247	7.77	<.0001
time	1.2475	0.05994	998	20.81	<.0001
A*time	0.7057	0.08581	998	8.22	<.0001
age	0.08860	0.01185	247	7.48	<.0001

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
A	1	247	60.33	<.0001
time	1	998	433.11	<.0001
A*time	1	998	67.63	<.0001
age	1	247	55.92	<.0001

Estimates					
Label	Estimate	Standard Error	DF	t Value	Pr > t
Effect of A at time=0	2.2148	0.2851	247	7.77	<.0001
Effect of A at time=4	5.0375	0.2851	998	17.67	<.0001

```

/*-----*
|  Model 2: Add precision variable Z (not confounder if unrelated to A)
*-----*/
title2 "Model 2: + age + z (precision covariate)";
proc mixed data=demo_long method=ml;
  class id;
  model y = A time A*time age z / solution;
  repeated / subject=id type=cs;

  estimate "Effect of A at time=0" A 1;
  estimate "Effect of A at time=4" A 1 A*time 4;
run;

```

Model 2: + age + z (precision covariate)

The Mixed Procedure

Model Information	
Data Set	WORK.DEMO_LONG
Dependent Variable	y
Covariance Structure	Compound Symmetry
Subject Effect	id
Estimation Method	ML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information

Class	Levels	Values
id	250	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250

Dimensions	
Covariance Parameters	2
Columns in X	6
Columns in Z	0
Subjects	250
Max Obs per Subject	5

Number of Observations	
Number of Observations Read	1250
Number of Observations Used	1250
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Log Like	Criterion
0	1	5907.93521697	
1	1	5744.36514538	0.00000000

Convergence criteria met.

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
CS	id	2.0098
Residual		4.5994

Fit Statistics	
-2 Log Likelihood	5744.4
AIC (Smaller is Better)	5760.4

Fit Statistics		
AICC (Smaller is Better)		5760.5
BIC (Smaller is Better)		5788.5
Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
1	163.57	<.0001

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	19.5695	0.5877	246	33.30	<.0001
A	2.2735	0.2808	246	8.10	<.0001
time	1.2475	0.05994	998	20.81	<.0001
A*time	0.7057	0.08581	998	8.22	<.0001
age	0.08862	0.01154	246	7.68	<.0001
z	0.3933	0.1065	246	3.69	0.0003

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
A	1	246	65.53	<.0001
time	1	998	433.11	<.0001
A*time	1	998	67.63	<.0001
age	1	246	58.99	<.0001
z	1	246	13.63	0.0003

Estimates					
Label	Estimate	Standard Error	DF	t Value	Pr > t
Effect of A at time=0	2.2735	0.2808	246	8.10	<.0001
Effect of A at time=4	5.0962	0.2808	998	18.15	<.0001

```

/*-----*
|  Model 3: Add time-varying biomarker (possible mediator / over-adjustment)
|  Use this to SHOW attenuation and discuss why we might exclude it
|  if the goal is the total effect of A.
|-----*/
title2 "Model 3: + age + z + biom (potential over-adjustment)";
proc mixed data=demo_long method=ml;
  class id;
  model y = A time A*time age z biom / solution;
  repeated / subject=id type=cs;

  estimate "Effect of A at time=0" A 1;
  estimate "Effect of A at time=4" A 1 A*time 4;
run;

title2;

```

Model 3: + age + z + biom (potential over-adjustment) The Mixed Procedure

Model Information		
Data Set	WORK.DEMO_LONG	
Dependent Variable	y	
Covariance Structure	Compound Symmetry	
Subject Effect	id	
Estimation Method	ML	
Residual Variance Method	Profile	
Fixed Effects SE Method	Model-Based	
Degrees of Freedom Method	Between-Within	
Class Level Information		
Class	Levels	Values
id	250	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203

Class Level Information

Class	Levels	Values
		204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250

Dimensions

Covariance Parameters	2
Columns in X	7
Columns in Z	0
Subjects	250
Max Obs per Subject	5

Number of Observations

Number of Observations Read	1250
Number of Observations Used	1250
Number of Observations Not Used	0

Iteration History

Iteration	Evaluations	-2 Log Like	Criterion
0	1	5723.48229637	
1	2	5539.19842734	0.00000000

Convergence criteria met.

Covariance Parameter Estimates

Cov Parm	Subject	Estimate
CS	id	1.8556

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
Residual		3.8503

Fit Statistics	
-2 Log Likelihood	5539.2
AIC (Smaller is Better)	5557.2
AICC (Smaller is Better)	5557.3
BIC (Smaller is Better)	5588.9

Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
1	184.28	<.0001

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	19.7437	0.5557	246	35.53	<.0001
A	1.0398	0.2752	246	3.78	0.0002
time	1.1856	0.05500	997	21.56	<.0001
A*time	0.5119	0.07957	997	6.43	<.0001
age	0.08761	0.01092	246	8.02	<.0001
z	0.3164	0.1010	246	3.13	0.0019
biom	1.0760	0.07197	997	14.95	<.0001

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
A	1	246	14.27	0.0002
time	1	997	464.63	<.0001
A*time	1	997	41.38	<.0001
age	1	246	64.33	<.0001
z	1	246	9.82	0.0019
biom	1	997	223.51	<.0001

Estimates					
Label	Estimate	Standard Error	DF	t Value	Pr > t
Effect of A at time=0	1.0398	0.2752	246	3.78	0.0002
Effect of A at time=4	3.0874	0.2950	997	10.47	<.0001

```

/* Create subject-specific baseline biomarker biom0 (value at time=0) */
proc sort data=demo_long;
  by id time;
run;

data demo_with_biom0;
  set demo_long;
  by id;
  retain biom0;
  if first.id then biom0 = .;
  if time = 0 then biom0 = biom; /* store baseline biomarker */
  /* carry biom0 forward to later rows for the same subject */
run;

/* Fill biom0 down within each id (in case time=0 isn't first, or missing) */
proc sort data=demo_with_biom0;
  by id time;
run;

data demo_with_biom0;
  set demo_with_biom0;
  by id;
  retain biom0_hold;
  if first.id then biom0_hold = biom0;
  else if missing(biom0_hold) and not missing(biom0) then biom0_hold = biom0;
  else if not missing(biom0) then biom0_hold = biom0;
  biom0 = biom0_hold;
  drop biom0_hold;
run;

proc print data = demo_with_biom0 (obs=20);
run;

```

Obs	id	age	A	z	time	biom	y	biom0
1	1	51.1130	1	-1.77231	0	-0.53577	24.3368	-0.53577
2	1	51.1130	1	-1.77231	1	-1.23809	25.7489	-0.53577
3	1	51.1130	1	-1.77231	2	0.11968	24.8903	-0.53577
4	1	51.1130	1	-1.77231	3	0.47853	32.5064	-0.53577
5	1	51.1130	1	-1.77231	4	1.71218	30.9145	-0.53577
6	2	38.0998	1	0.61437	0	0.73368	25.7812	0.73368
7	2	38.0998	1	0.61437	1	1.35669	30.8212	0.73368

Obs	id	age	A	z	time	biom	y	biom0
8	2	38.0998	1	0.61437	2	2.47425	33.4281	0.73368
9	2	38.0998	1	0.61437	3	0.18479	31.7852	0.73368
10	2	38.0998	1	0.61437	4	0.08670	30.5793	0.73368
11	3	57.4642	1	-0.35376	0	0.40352	24.5160	0.40352
12	3	57.4642	1	-0.35376	1	2.62107	28.4818	0.40352
13	3	57.4642	1	-0.35376	2	3.25722	32.2862	0.40352
14	3	57.4642	1	-0.35376	3	3.61449	35.8133	0.40352
15	3	57.4642	1	-0.35376	4	4.02203	34.3625	0.40352
16	4	55.1957	0	-2.13380	0	0.07050	22.9312	0.07050
17	4	55.1957	0	-2.13380	1	-1.14543	20.0592	0.07050
18	4	55.1957	0	-2.13380	2	-0.21004	22.3938	0.07050
19	4	55.1957	0	-2.13380	3	0.58952	26.3002	0.07050
20	4	55.1957	0	-2.13380	4	-0.42391	23.2728	0.07050

```
/* Fit model: baseline biomarker adjusted, but NOT time-varying biom */
title "Model: adjust for baseline biomarker biom0 (not time-varying biom)";
proc mixed data=demo_with_biom0 method=ml;
  class id;
  model y = A time A*time age z biom0 / solution;
  repeated / subject=id type=cs;

  /* Optional: exposure effect at time=0 and time=4 */
  estimate "Effect of A at time=0" A 1;
  estimate "Effect of A at time=4" A 1 A*time 4;
run;
title;
```

Model: adjust for baseline biomarker biom0 (not time-varying biom)
The Mixed Procedure

Model Information	
Data Set	WORK.DEMO_WITH_BIOM0
Dependent Variable	y
Covariance Structure	Compound Symmetry
Subject Effect	id
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information		
Class	Levels	Values
id	250	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250

Dimensions	
Covariance Parameters	2
Columns in X	7
Columns in Z	0
Subjects	250
Max Obs per Subject	5

Number of Observations	
Number of Observations Read	1250
Number of Observations Used	1250
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	5921.57939561	
1	1	5761.77129244	0.00000000

Convergence criteria met.

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
CS	id	1.9989
Residual		4.6087

Fit Statistics	
-2 Res Log Likelihood	5761.8
AIC (Smaller is Better)	5765.8

Fit Statistics		
AICC (Smaller is Better)		5765.8
BIC (Smaller is Better)		5772.8
Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
1	159.81	<.0001

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	19.6394	0.5876	245	33.43	<.0001
A	1.9111	0.3186	245	6.00	<.0001
time	1.2475	0.06000	998	20.79	<.0001
A*time	0.7057	0.08590	998	8.22	<.0001
age	0.08857	0.01152	245	7.69	<.0001
z	0.3854	0.1064	245	3.62	0.0004
biom0	0.3387	0.1409	245	2.40	0.0170

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
A	1	245	35.98	<.0001
time	1	998	432.25	<.0001
A*time	1	998	67.50	<.0001
age	1	245	59.11	<.0001
z	1	245	13.11	0.0004

Type 3 Tests of Fixed Effects						
Effect	Num DF	Den DF	F Value	Pr > F		
biom0	1	245	5.78	0.0170		
Estimates						
Label		Estimate	Standard Error	DF	t Value	Pr > t
Effect of A at time=0		1.9111	0.3186	245	6.00	<.0001
Effect of A at time=4		4.7338	0.3186	998	14.86	<.0001

```
title "Final Model: adjust for baseline biomarker biom0 (not time-varying biom)";
proc mixed data=demo_with_biom0;
  class id;
  model y = A time A*time age z biom0 / solution;
  repeated / subject=id type=cs;

  /* Optional: exposure effect at time=0 and time=4 */
  estimate "Effect of A at time=0" A 1;
  estimate "Effect of A at time=4" A 1 A*time 4;
run;
title;
```

Final Model: adjust for baseline biomarker biom0 (not time-varying biom)

The Mixed Procedure

Model Information	
Data Set	WORK.DEMO_WITH_BIOM0
Dependent Variable	y
Covariance Structure	Compound Symmetry
Subject Effect	id
Estimation Method	REML
Residual Variance Method	Profile
Fixed Effects SE Method	Model-Based
Degrees of Freedom Method	Between-Within

Class Level Information		
Class	Levels	Values
id	250	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250

Dimensions	
Covariance Parameters	2
Columns in X	7
Columns in Z	0
Subjects	250
Max Obs per Subject	5

Number of Observations	
Number of Observations Read	1250
Number of Observations Used	1250
Number of Observations Not Used	0

Iteration History			
Iteration	Evaluations	-2 Res Log Like	Criterion
0	1	5921.57939561	
1	1	5761.77129244	0.00000000

Convergence criteria met.

Covariance Parameter Estimates		
Cov Parm	Subject	Estimate
CS	id	1.9989
Residual		4.6087

Fit Statistics	
-2 Res Log Likelihood	5761.8
AIC (Smaller is Better)	5765.8

Fit Statistics		
AICC (Smaller is Better)		5765.8
BIC (Smaller is Better)		5772.8
Null Model Likelihood Ratio Test		
DF	Chi-Square	Pr > ChiSq
1	159.81	<.0001

Solution for Fixed Effects					
Effect	Estimate	Standard Error	DF	t Value	Pr > t
Intercept	19.6394	0.5876	245	33.43	<.0001
A	1.9111	0.3186	245	6.00	<.0001
time	1.2475	0.06000	998	20.79	<.0001
A*time	0.7057	0.08590	998	8.22	<.0001
age	0.08857	0.01152	245	7.69	<.0001
z	0.3854	0.1064	245	3.62	0.0004
biom0	0.3387	0.1409	245	2.40	0.0170

Type 3 Tests of Fixed Effects				
Effect	Num DF	Den DF	F Value	Pr > F
A	1	245	35.98	<.0001
time	1	998	432.25	<.0001
A*time	1	998	67.50	<.0001
age	1	245	59.11	<.0001
z	1	245	13.11	0.0004

Type 3 Tests of Fixed Effects						
Effect	Num DF	Den DF	F Value	Pr > F		
biom0	1	245	5.78	0.0170		
Estimates						
Label		Estimate	Standard Error	DF	t Value	Pr > t
Effect of A at time=0		1.9111	0.3186	245	6.00	<.0001
Effect of A at time=4		4.7338	0.3186	998	14.86	<.0001